
UNIT 6 REAL BUSINESS CYCLES*

Structure

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12.0 OBJECTIVES

After going through this unit you would be in a position to

- describe the salient features of new-classical economics;
- outline the assumptions and methodology adopted by new-classical economics on analysis of business cycles;
- bring out the important features of real business cycle theory;
- explain the inter-temporal substitution in labour supply;
- assess the role of supply shocks; and
- discuss the new-Keynesian approach to business cycle.

12.1 INTRODUCTION

In the previous Unit we mentioned that business cycles are synchronic in the sense that the state of recession or boom is prevalent in most industries at the same time. Further, certain variables are pro-cyclical while some others are counter-cyclical. Examples of pro-cyclical variables are output, employment, wages and consumption expenditure. Examples of counter-cyclical variables are unemployment, company profits, and stock market prices.

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There is no consensus among economists about the reasons behind business cycles. According to Keynes (1936) fluctuations in investment (and therefore employment) is caused by ‘animal spirit’, a sort of human emotion. Fluctuation in output gets amplified due to the effect of ‘investment multiplier’. Samuelson (1939) showed that the interaction of multiplier and accelerator leads to business cycles. According to Austrian economist Joseph A. Schumpeter (1961), business cycles are an outcome of the process of innovation and technological change; a process called ‘creative destruction’. Schumpeter emphasises on the role of entrepreneurs, who undertake innovation – they bring in new technology which disrupts existing production processes. Such disruptions in an economy lead to substantial fluctuations in employment and output. There have been several other theories of business cycles which we have not included in the previous Unit. Richard Goodwin (1967), another heterodox economist, ascribes business cycles to the fluctuations in the share of workers in total output. During the boom phase of business cycle, rising employment leads to higher wages, rise in bargaining power of workers, and increase in the share of wages in revenue. This leads to decrease in profit share, decrease in accumulation of capital, and consequent decline in employment. Eventually the rise in unemployment leads to fall in the share of wages in revenue, consequent recovery of profits, and increase in accumulation. In sharp contrast to Keynes, Milton Friedman considered the changes in money supply to be the major reason behind business cycles. According to Friedman, a free market economy is inherently stable. The fluctuation in money supply causes business cycles in an economy.

New classical economists followed the monetarist tradition and emphasized on monetary factors. They introduced rational expectations and microfoundation in macroeconomic analysis. According to them monetary shocks causes business cycles in an economy. The real business cycle theory, however, brings back the importance of real factors such as productivity shocks to explain business cycle.

In this Unit we begin with new-classical view on business cycle and subsequently move on to real business cycle theory. We will conclude the Unit with a note on new-Keynesian view on business cycle.

12.2 NEW-CLASSICAL VIEW ON BUSINESS CYCLE

During the 1950s and 1960s Keynesian economics was very popular – its policy prescription was acceptable to researchers as well as policy makers. Milton Friedman, however, challenged Keynesian economics during that time. According to Friedman, changes in aggregate income are explained mostly by monetary policy. Had the government taken recourse to open market operations and redeemed bonds and securities during the Great Depression, money supply would have increased and the adverse impact would have been lesser. Further, Friedman disagreed with the inflation-unemployment trade-off as suggested by the Phillips Curve. He introduced expectations into the analysis, and pointed out that the Phillips curve is vertical in the short-run. The trade-off is possible only in

the short-run; in the long-run, unemployment cannot be brought down below the natural rate of unemployment.

The onset of stagflation during the 1970s further consolidated the dissent on Keynesian economics. New-classical economics, led by Robert Lucas and Thomas Sargent, brought in a revolution in macroeconomics. New-classical economists revived the classical ideas; the analytical tools however were different. Thus, new-classical economics is completely opposed to the Keynesian policy prescriptions.

New-classical economics is based on three assumptions: (i) rational expectations hypothesis, (ii) continuous market clearing, and (iii) the aggregate supply hypothesis. People have rational expectations; they can correctly anticipate the effect of government actions; and they take counter-measures to neutralise the impact of government policy. A major inference drawn by new-classical economics is that policy measures taken by the government are ineffective. Only unexpected policy shocks lead to temporary deviations in output and employment from their natural levels. The second assumption states that markets clear at any point of time. An implication of the above is the equality between demand and supply, and the economy is in a state of equilibrium always. New-classical economics assumes flexibility in wage rate and prices. Further, adjustments in prices and wages are instantaneous. You may be wondering how the labour market can clear (i.e., there is no unemployment)! This in fact is a debatable assumption of new-classical economics – according to new-classical economists, unemployment is a voluntary phenomenon. At the ongoing wage rate anybody who is willing to work gets employed. The third assumption states that the supply of labour by workers and the supply of output of firms depend upon relative prices. This is the famous Lucas supply function discussed in Unit 7.

Three prominent features of new-classical economics are as follows: (i) microfoundation, (ii) dynamic stochastic general equilibrium, and (iii) quantitative calibration of the economy. Microfoundation means that macroeconomic phenomena are explained through the behaviour of microeconomic agents such as households and firms. For example, the decision of a household is based on its utility function while that of a firm is based on a production function. New-classical models are based on dynamic stochastic general equilibrium (DSGE). As the name suggests, these are ‘general equilibrium models’ in the sense that they pertain to the whole economy. The word ‘dynamic’ reflects the idea that these are inter-temporal models and economic agents optimise on their decision-variables over time. Thus, DSGE models involve the issue of ‘dynamic optimisation’ (See MEC-203, Block 7). As you know, the economy receives various types of shocks, viz., demand shocks, supply shocks and policy shocks, which are exogenous. These shocks are assumed to be random in nature; thus these models are ‘stochastic’. In simple DSGE models there are three inter-related blocks (set of equations), viz., (i) a

demand block, (ii) a supply block, and (iii) a monetary policy equation (for example, policy rule). The equations included in these blocks are derived from micro-foundations.

In order to describe the economy, new-classical economists use the method of calibration. The DSGE models require solution to a set of inter-related dynamic optimisation problems. Here households and firms make decisions on the basis of current and expected future paths for prices and policy variables. This is a challenging task as data are not easily available for longer time periods. Therefore, in DSGE models researchers make extensive use of numerical iteration, so as to replicate the economy. The researcher chooses (from past studies) a set of parameters to describe the utility function and production function. In the process, they rely on model simulation and take parameter values that best fit the empirical data. Such a procedure of comparing the model output with empirical data is called calibration. Remember that calibration is different from estimation used in econometrics.

A question arises at this point: Why do employment and output fluctuate in an economy, if there is market-clearing all the time? In other words, why do business cycles occur? According to new-classical economics, there are two basic reasons: (i) inter-temporal substitution between labour and leisure, and (ii) Lucas supply function. If the actual real wage rate is higher than the expected real wage rate, the household substitutes leisure by labour – thereby working for higher number of hours in a day. In the process it supplies more labour. Thus, there is an increase in supply of labour, which leads to an increase in output (higher than the potential output). As pointed out above, changes in employment are treated as voluntary choices of workers – the amount of labour supplied by households depends upon their rational expectations about real wages. The ‘Lucas Supply Function’ suggests that firms suffer from signal extraction problem and mistake a general price rise as a relative price rise. When there is a general price rise, there is no relative increase in prices of the products produced by a firm, as the cost of production also has gone up. However, firms often see that there is an increase in the prices of their products and they increase their production. Thus, there is a (positive) deviation in employment and output from their natural levels. This leads to a boom or expansion phase in the economy.

Let us take the opposite scenario. Suppose, households perceive that the actual wages they receive is less than expected wage. As a response, households will work less (enjoy more leisure) and supply less labour to the market. Decrease in supply of labour will result in a (negative) deviation in output from its natural level. If the firms perceive that there is a relative decrease in prices of their products, they will decrease their output level. Such decline in output will lead to a recession in the economy. Thus, while the economy is in equilibrium, there could be fluctuation in employment and output.

Check Your Progress 1

- 1) What are the basic assumptions of new-classical economics?

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- 2) Explain the reasons put forth by new-classical economists for occurrence of business cycles in an economy.

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12.3 REAL FACTORS VS. MONETARY FACTORS

Real business cycle (RBC) theory originated with the publication of a paper in 1982 by Kydland and Prescott entitled, “Time to Build and Aggregate Fluctuations”. Kydland and Prescott stated that business cycle is not due to fluctuation in monetary variables; rather it is caused by real shocks to the economy. The RBC theory is a distinct class of new-classical macroeconomics. Therefore, the basic assumptions and analytical tools of new-classical models are applied in RBC models also. In fact, Kydland and Prescott were the first to characterise the general equilibrium of a full-fledged dynamic and stochastic macroeconomic model.

There could be two types of shocks to the economy: (i) nominal shocks, and (ii) real shocks. The nominal shocks (shocks to money supply or money demand) affect the LM curve. The real shocks (shocks to the production function, real government spending, or to savings and consumption decisions) affect the IS curve. In other words, real shocks affect the goods market equilibrium and the labour market equilibrium.

As the name suggests, the RBC models ascribe ‘real shocks’ as the prime reason behind business cycles. The shocks to the production function are known as supply shocks or productivity shocks. Supply shocks can arise due to the following reasons:

- (i) development of new products or production methods (production technology)
- (ii) changes in the quality of labour or capital
- (iii) changes in the availability of raw materials (relative prices, oil crisis)
- (iv) unusually good or bad weather (drought, flood, earthquake)

- (v) changes in government regulations affecting production (tax rate, subsidies, etc.)

The positive or beneficial productivity shocks result in economic booms, whereas the negative or adverse productivity shocks cause recessions in an economy. Let us discuss the effect of a temporary adverse supply shock on the economy (see Fig. 12.1). There will be a decline in the marginal productivity of labour, which will reduce the demand for labour. Consequently, there will be a fall in the real wage. Households will see that their real wage has declined and they will supply lower quantity of labour. This will lead to a fall in the level of output (consequently, a recession). A beneficial productivity shock, on the other hand, would result in an economic boom. There would be temporary hike in productivity in the economy. This will cause an increase in investment and work effort in the current period. Consequently, both the real output and consumption spending will be increased. Once the effect of the supply shock dies down, the economy will be back to natural rate of employment and output.

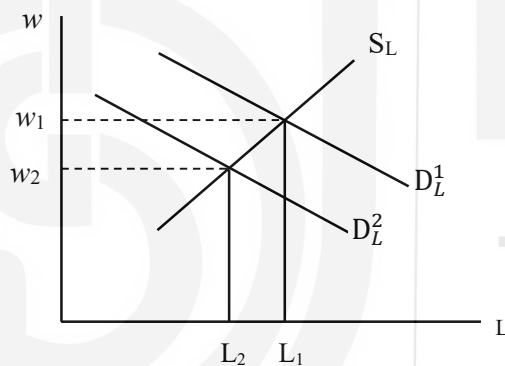


Fig. 12.1: Effect of Temporary Adverse Supply Shock

The RBC theory is based on two basic principles: (i) money has little role to play in business cycles, and (ii) business cycles occur as a response of rational economic agents (firms and households) to real shocks. The economy is continuously at equilibrium – the supply of labour and output are according to rational expectations of economic agents.

The RBC theory has certain differences with new-classical economics. There are at least three differences between the real business cycle theory and the new classical school of thought: (i) the RBC theory does not distinguish between short-run and long-run. It integrates theory of economic growth with business cycle theory. (ii) Lucas advocated that households and firms get fooled – they cannot extract correct signal from price changes in the economy. This results in deviation of actual output from potential output. The RBC theory assumes that households and firms have complete information. (iii) RBC theory assumes that supply shocks arise from real factors while new-classical economics assumes that shocks arise from monetary factors.

The implications of RBC theory are as follows: (i) The government should not try to stabilise output level; it should try to stabilise prices (i.e., inflation should be kept under control – inflation targeting). (ii) The government should not influence aggregate demand through fiscal policy measures (taxes and subsidies), as fiscal measures may not be effective (due to the issue of Ricardian equivalence). (iii) The government should use fiscal policy to stimulate the supply side of the economy (to provide positive supply shocks to the economy).

Real business cycle is criticised on certain grounds: Its assumption of instantaneous market clearing is unrealistic. Prices and wages are rigid in an economy. Further, there is little evidence of large aggregate technology shocks in an economy.

12.4 A BASELINE RBC MODEL

Let us discuss how the fluctuations of aggregated macroeconomic variables can be explained through a general equilibrium model. Recall from microeconomic theory that the Walrasian model (a baseline general equilibrium model) is a competitive model without any of the following: (i) asymmetric information, (ii) externalities, (iii) missing markets, and (iv) other imperfections. If we can explain macroeconomic fluctuations through a Walrasian model, there is no fundamental departure from conventional microeconomic analysis. Real-business-cycle theory aims to analyse how a Walrasian model provides a good explanation of the nature and character of observed fluctuations.

The Ramsey model (see Unit 8) can be seen as a baseline general equilibrium model. The model excludes market imperfections; and there is no heterogeneity among households. It is based on microfoundation in the sense that it considers a representative household and a representative firm. The representative household consumes goods and services which are homogenous. Goods and services produced in the economy are used for consumption by households as well as investment by firms. The representative household supplies labour to the firm. The household determines the quantity of labour to be supplied on the basis of a utility function. Recall that in the Ramsey model an economy converges to a balanced growth path and then grows smoothly.

Let us try to extend the Ramsey model to include aggregate macroeconomic fluctuations. In the absence of shocks, the economy is on a steady state. Therefore, we need to incorporate certain disturbance to the model.

Suppose that there is certain disturbance or shock to the economy. What will be the impact of such shocks? Will the impact of the shock dissipate (die down) after some periods or will it generate fluctuations in output in the economy? The sources of shocks to an economy could be of two types: (i) there could be a shock to the economy's technology. The production technology of the economy undergoes change. Such shocks will lead to a change in the production function across periods. (ii) Change in government purchases also could be a source of disturbance to the economy. These two types of shocks are real disturbances; they are not monetary or nominal disturbances. The change in technology leads

to change in the amount of output produced from a given quantity of inputs. Similarly, the change in government-purchases leads to change in the quantity of goods that are available to the private economy for a given level of production. Since the shocks to an economy are real factors, the models are known as *real-business-cycle* (RBC) models.

We need to make a second modification in the Ramsey model – we allow for change in employment. In the traditional business cycle theories, labour supply is assumed to be exogenous; labour supply is either constant or growing smoothly. As mentioned above, the RBC models are based on microfoundation such that labour supply is determined by a utility function. Households allocate their time between work and leisure – an increase in the time allocated to work implies an increase in the supply of labour and a decrease in the quantity of leisure.

Let us assume that the economy consists of a large number of homogenous price-taking firms and households. There are two inputs to the production function, viz., capital (K) and labour (L). Let us assume that technology is exogenous and it is given by A_t for time period ' t '. Further, let us assume that the functional form of the production function is Cobb-Douglas.

Thus, output in period t is

$$Y_t = A_t F(K_t, L_t) = K_t^\alpha (A_t L_t)^{1-\alpha} \quad \dots (12.1)$$

where $0 < \alpha < 1$

We know that output is divided among consumption (C), investment (I), and government purchases (G). If δ is the depreciation rate of capital in each period, the capital stock in period $(t + 1)$ is

$$\begin{aligned} K_{t+1} &= K_t + I_t - \delta K_t \\ &= K_t + Y_t - C_t - G_t - \delta K_t \end{aligned} \quad \dots (12.2)$$

(Since $C_t + I_t + G_t = Y_t$; we have $I_t = Y_t - C_t - G_t$)

Let us consider that labour and capital are paid their marginal products. Thus, the real wage and the real interest rate in period t are

$$\begin{aligned} w_t &= (1 - \alpha) A_t K_t^\alpha (A_t L_t)^{-\alpha} \\ &= (1 - \alpha) A_t (K_t / A_t L_t)^{-\alpha} \end{aligned} \quad \dots (12.3)$$

and,

$$r_t = \alpha (A_t L_t / K_t)^{1-\alpha} \quad \dots (12.4)$$

(You can derive equation (12.3) and equation (12.4) from equation (12.1) by taking partial derivatives with respect to L and K respectively)

The representative household maximizes the expected value of the utility function

$$U = \sum_{t=0}^{\infty} e^{-\rho t} u(c_t, 1 - l_t) \frac{N_t}{H} \quad \dots (12.5)$$

where, $u(\bullet)$ is the instantaneous utility function. Here, ρ is the discount rate, which represents time preference of utility (i.e., the rate at which future income is discounted). Further, N_t is population and H is the number of households. Then, the average number of members in each household is N_t/H . If the exogenous growth rate of population is n :

$$\ln N_t = \bar{N} + nt, \quad \dots (12.6)$$

where $n < \rho$

Thus, the level of N_t is given by $N_t = e^{\bar{N}+nt}$

Notice that $u(\bullet)$ has two factors in equation (12.5): (i) consumption per member of the household (c_t), and (ii) leisure per member (l_t). Leisure is the difference between the time endowment per member (normalized to 1 for simplicity) and the amount each member works. As all households are homogenous, $c = C/N$, and $l = L/N$.

Let us first discuss about the assumption on the technology shocks. To capture trend growth, we assume the following functional form for technology:

$$\ln A_t = \bar{A} + gt \quad \dots (12.7)$$

where g is the rate of technological progress. Technology in the above specification is not subjected to random shocks. In order to include random disturbances in technology we consider the following functional form:

$$\ln A_t = \bar{A} + gt + \tilde{A}_t \quad \dots (12.8)$$

where $\tilde{A}$ is the random disturbances. The term $\tilde{A}_t$ follows first-order autoregressive process. That is,

$$\tilde{A}_t = \rho_A \tilde{A}_{t-1} + \varepsilon_{A,t}, \quad -1 < \rho_A < 1 \quad \dots (12.9)$$

where, the $\varepsilon_{A,t}$ is a stochastic error term (white noise). It means that $\varepsilon_{A,t}$ has zero mean and zero correlation with other variables in the equation. Hence, equation (12.9) indicates that the random component $\tilde{A}_t$ is a fraction ρ_A of the previous period's value and a white noise error. The effect of a shock to technology disappears gradually over time.

We follow a similar functional form for government purchases: It has a trend component and a white-noise component. There is a trend growth rate of per capita government purchases, but over time government purchases may become arbitrarily large or arbitrarily small relative to the economy. Thus,

$$\ln G_t = \bar{G} + (n + g)t + \tilde{G}_t, \quad -1 < \rho_A < 1, \quad \dots (12.10)$$

$$\tilde{G}_t = \rho_G \tilde{G}_{t-1} + \varepsilon_{G,t}, \quad -1 < \rho_A < 1, \quad \dots (12.11)$$

where the $\varepsilon_{G,t}$ is white noise. This completes the description of the RBC model.

12.5 INTER-TEMPORAL SUBSTITUTION IN LABOUR SUPPLY

We noted in the previous section that major differences between RBC model and the Ramsey model are: (i) the inclusion of leisure in the utility function, and (ii) the introduction of randomness in technology and government purchases as the sources of macroeconomic aggregate fluctuations.

Let us focus on the introduction of leisure in the utility function. For simplicity assume that the household lives only for one period and has only one member, which has no initial wealth. Now, the objective function of the household is $\ln c + b \ln(1 - l)$ and its budget constraint is $c = wl$. Hence, the Lagrangian maximization problem is:

$$\zeta = \ln c + b \ln(1 - l) + \lambda(wl - c) \quad \dots (12.12)$$

The first-order conditions for c and l are as follows:

$$\frac{1}{c} - \lambda = 0, \quad \dots (12.13)$$

$$-\frac{b}{1-l} + \lambda w = 0, \quad \dots (12.14)$$

From equations (12.13), we find that $\lambda = \frac{1}{c} = \frac{1}{wl}$ (since $c = wl$ as per the budget constraint).

Now, substituting the value of λ in equation (12.14), we obtain

$$-\frac{b}{1-l} + \frac{1}{wl} w = 0$$

That is,

$$-\frac{b}{1-l} + \frac{1}{l} = 0 \quad \dots (12.15)$$

The labour supply that satisfies equation (12.15) is independent of the wage rate as wage rate is not included in this equation. The reason is that utility is logarithmic in consumption and the household has no initial wealth. The income effect and substitution effect of a change in the wage rate offset each other. This is the static case, as we assumed that the household lives for one period.

If we consider the case where the household lives for two periods or more than two periods, then the wage variations do affect labour supply. Let us continue to assume that household has no initial wealth with one member who lives for two periods. Also, assume that there is no uncertainty about the interest rate or the second-period wage.

The household's lifetime budget constraint is now

$$c_1 + \frac{1}{1+r} c_2 = w_1 l_1 + \frac{1}{1+r} w_2 l_2 \quad \dots (12.16)$$

where r is the real interest rate. The Lagrangian maximization problem is:

$$\zeta = \ln c_1 + b \ln(1 - l_1) + e^{-\rho} [\ln c_2 + b \ln(1 - l_2)]$$